The degree sixteen boundary

Human Compression project: a component of the L044 recovery

Put $V = \mathbb{F}_2^3$, and identify sixteen points with $V \times \mathbb{F}_2$. The binary partition argument previously established for L044 leads to the following permutation group:

$$W = \{(a,b) \mapsto (a+u, b+v(a)) : u \in V, \quad v : V \to \mathbb{F}_2, \quad \sum_{a \in V} v(a) = 0\}. \tag{1}$$

We resolve its remaining finite obligation. All group operations in the certificate are exact permutation operations in GAP. The proof of the reduction to (1) is acquired separately; it is not replaced by this computation.

Theorem 1. *No transitive subgroup of W has a string C-marking of rank five. Disconnected strings, including any inherited five-label subset of a longer string, are included.*

Proof. We give the exhaustive algorithm and its output. A marking $(x_i)_{i \in I}$ has the intersection property if

$$\langle x_i : i \in J \rangle \cap \langle x_i : i \in K \rangle = \langle x_i : i \in J \cap K \rangle \qquad (J, K \subseteq I). \tag{2}$$

Thus every generator is a nonidentity involution outside the subgroup generated by the others. Every subfamily also satisfies (2). In a string, only consecutive generators can fail to commute. Separate its noncommuting components, and reorder the components by decreasing size. Reordering does not change the generated group or (2), and distinct components commute. Within a component we retain its linear order. Consequently the possible component sizes for five generators are

$$5, \quad 4+1, \quad 3+2, \quad 3+1+1, \quad 2+2+1, \quad 2+1+1+1, \quad 1+1+1+1+1. \tag{3}$$

The last case is impossible: (2) gives an elementary abelian group of order 32. A faithful transitive action of an abelian group is regular, since an element fixing one point fixes all its translates. It cannot have degree sixteen. The six other cases are enumerated.

There are 2^7 even functions v, and eight translations, so $|W| = 1024$. A nonidentity element with $u = 0$ is an involution, giving 127 such elements. For each $u \neq 0$, the involution condition is $v(a+u) = v(a)$; the four pairs of the translation allow 16 functions, all of even weight. There are therefore $127 + 7 \cdot 16 = 239$ involutions, precisely the initial search domain.

For orbit reduction take the permutation group N generated by all eight individual pair flips, the three coordinate translations, a cyclic permutation of the coordinates, and the transvection $(a_0, a_1, a_2) \mapsto (a_0 + a_1, a_1, a_2)$. Each generator normalizes W. Affine changes of a preserve even weight; conjugation by a flip function w adds $w(a) + w(a+u)$ to v, whose sum is zero. The supplied generators have order 344064 as a group, checked directly in the model. The argument needs only normalization, not an assertion that N is the full normalizer or automorphism group.

The six label sets used are the rows of the table below. Gaps separate components. In each row the labels are visited in positions $1, 5, 2, 4, 3$. Suppose the current chosen labels are S, their generated subgroup is H, and their pointwise stabilizer under conjugation by N is $C = C_N(x_i : i \in S)$. For the next label j, form the complete set of the 239 involutions $x \in W$ satisfying

$$x \notin H, \qquad x x_i = x_i x \iff |i - j| > 1 \quad (i \in S). \tag{4}$$

This set is invariant under C. The program forms its actual C-orbits, checks their disjoint union against the entire candidate set, and takes one representative from each. It retains a representative exactly when the enlarged family satisfies (2). Recursion replaces H by the enlarged subgroup and C by $C_C(x)$. At depth five it tests actual transitivity on all sixteen points. No requirement that the subgroup equal W is imposed.

This recursion is exhaustive by induction on the depth. Any qualifying full family has a retained restriction by (2); its next member lies in (4). Conjugating by the current stabilizer sends that member to one of the representatives while leaving the already chosen family fixed. Such conjugation preserves all remaining conditions. Hence some complete branch represents the full family. Neither subgroup order targets nor time caps are used to infer a completed search.

For every proposed extension, the reference routine explicitly constructs the subgroups for all subsets of the chosen family. Their sets of elements are bit vectors in the fixed list of all 1024 elements of W. For every unordered pair of subsets it compares their bit-vector intersection with the subgroup for the intersection of the label subsets. This is exactly (2), including the empty subset. A second routine uses the acquired recursive end-intersection criterion on each consecutive component and checks that their commuting product has order equal to the product of their orders. The two decisions are compared before every pruning step. A disagreement aborts the computation. Thus even rejected extensions are checked against (2); the reference decision is not restricted to the eventual accepted leaves.

Sizes	Labels	Nodes at depths $0, \ldots, 5$	Leaves	Transitive
5	$1,2,3,4,5$	$1,6,76,76,32,0$	0	0
$4+1$	$1,2,3,4,6$	$1,6,76,76,528,0$	0	0
$3+2$	$1,2,3,5,6$	$1,6,76,76,32,0$	0	0
$3+1+1$	$1,2,3,5,7$	$1,6,76,76,528,5376$	5376	0
$2+2+1$	$1,2,4,5,7$	$1,6,76,76,528,0$	0	0
$2+1+1+1$	$1,2,4,6,8$	$1,6,76,76,528,2912$	2912	0

All six searches terminate. The reference routine is called 20478 times and performs 5413998 subset-pair comparisons. None of the 8288 complete families is transitive. This excludes all cases in (3), proving the theorem. □

Corollary 2. *Let A_n in its natural action have a string C-marking of rank $r \geq n/2$, where $n \geq 16$. Suppose every transitive maximal omission preserves a partition into two blocks. Then every double omission is intransitive.*

Proof. The acquired binary partition theorem proves that a transitive double omission under these hypotheses forces $n = 16$, $r = 8$, and a transitive subgroup of (1) generated by five of the original distinguished generators. Its full intersection property is inherited. Theorem 1 excludes that subgroup. □

Replay and controls. The files `model.g`, `enumeration.g`, and `run.g` implement the argument above. Running `python3 check.py NEW_DIRECTORY` copies those files to a fresh directory and performs the full computation. It requires all completion markers, an empty error stream, the complete table, and unchanged source hashes. The node and time caps cause failure, never acceptance. The four commuting generators consisting of the coordinate translations and the simultaneous flip of all pairs give a positive control: their subgroup is regular of order sixteen and satisfies (2). Repeating one translation gives a negative control which must fail (2). Control comparisons are excluded from the reported census totals.

Attribution and scope. The recovered target is the transitive double-omission branch of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 3.4. Its binary partition strategy is acknowledged in the separately reviewed reduction. Here that reduction permits the stronger local exclusion of all transitive five-mark subgroups of W, without extending them to eight generators. Orbit enumeration, the intersection property, and the recursive end criterion are established tools; the latter is acquired from our reviewed F004 enumeration component. All subgroup calculations use GAP. This packet proves only the displayed conditional branch. The case where every double omission is intransitive, and the other branches supplying the binary hypothesis, remain separate. No whole-manuscript recovery, open-conjecture resolution, Lean verification, or numerical compression factor is claimed here.